

Species-Aware Review: Bacterial RNase E RNA Degradosome in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03018` "RNA degradation" (15 primary genes; 17 candidate genes) **Commissioned module:** Reusable bacterial RNase E-centered RNA degradosome **Date:** 2026-08-31 · **Iterations:** 1–2 of 3

1. Executive summary

The RNase E-centered RNA degradosome is **present and satisfiable** in *P. putida* KT2440, but the organism-specific composition **diverges from the *Escherichia coli* textbook model**, and the KEGG `ppu03018` bucket is **substantially broader than the commissioned module**.

- **Core scaffold + endonuclease is solid.** RNase E (`rne`, PP_1905) is present, essential, and — as in all *Pseudomonadota* — nucleates the degradosome through its C-terminal domain (CTD). Direct *Pseudomonas* evidence (from *P. aeruginosa*) shows the CTD binds the core partners **PNPase and RhlB** through short linear motifs (SLiMs) (`P 40096066`).
- **PNPase and a DEAD-box helicase are present** (`pnp` PP_4708; `rhlB` PP_1295 plus paralogues). Which helicase is the *bona fide* degradosome partner in *P. putida* is **ambiguous** — *Pseudomonads* can use **RhlE** rather than RhlB, and can recruit the hydrolytic exoribonuclease **RNase R** in place of/alongside PNPase (`P 15705581`).
- **Enolase membership is homology-inferred and uncertain** in this lineage; its glycolytic role is certain, its degradosome role in KT2440 is not demonstrated (contrast *E. coli*, `P 23927922`).

- **The bucket over-reaches.** `recQ` (DNA repair), `rho` (transcription termination), `ppk` / `ppkB` (polyphosphate), and `groEL` / `dnaK` (chaperones) are neighboring-process or over-propagated members that should be kept out of the degradosome-core module.
- **Direct KT2440 experimental evidence exists** for RNase E, PNPase, RNase R (deletion phenotypes;

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and for Hfq (Crc-Hfq catabolite repression;

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— the latter clarifies that Hfq's characterized role here is translational regulation, not degradosome scaffolding.

Bottom line: Mark the four degradosome-core steps **covered** (with a `candidate_uncertain` flag on the enolase step and on helicase/exonuclease paralog identity). Recommend `module_needs_revision` on the fixed RhlB/PNPase wiring, `not_expected_in_target_taxon` / wrong-process for `recQ`, and separation of the 5'-trigger, 3'-decay, chaperone and polyphosphate members into neighboring modules.

2. Target-organism pathway definition

What the module includes (biochemical scope): the multi-protein machine that carries out bulk mRNA turnover and processing in Gram-negative bacteria — endonucleolytic initiation by RNase E, ATP-dependent unwinding of structured RNA by a DEAD-box helicase, and 3'→5' exonucleolytic degradation by PNPase — physically organized on the intrinsically disordered RNase E CTD scaffold, which in Pseudomonads also tethers the complex to the inner membrane

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Neighboring processes to keep separate (do NOT fold into the degradosome-core module): - **5'-end deprotection / decay initiation** — RppH (`rppH`, PP_5146): removes 5'-pyrophosphate to license RNase E cleavage; upstream trigger, not a core subunit

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- **3'-end-dependent decay & tailing** — RNase R (`rnr`, PP_4880) and poly(A) polymerase I (`pcnB`, PP_4697). - **Transcription termination** — Rho (`rho`, PP_5214). - **DNA metabolism** —

RecQ (`recQ`, PP_4516): a DNA helicase; not RNA degradation. - **Protein folding / stress** — GroEL (`groEL`, PP_1361), DnaK (`dnaK`, PP_4727): chaperones (primary bucket `ppu04156`); DnaK is at most a stress-induced minor degradosome associate in *E. coli* (`P 16139413`).

- **Polyphosphate metabolism** — PPK/PPK2 (`ppk` PP_5217, `ppkB` PP_0712): KEGG map03018 "others" box; peripheral. - **Glycolysis** — Enolase (`eno`, PP_1612): primary role in `ppu00010`.

Alternate names / database definitions: KEGG map03018 "RNA degradation" $\approx$ the "RNA degradosome" concept, but the KEGG map is deliberately inclusive (degradosome + exosome + accessory/others boxes). GO terms of interest: degradosome (GO:0030679 "cytoplasmic side"/complex terms), ribonuclease E activity (GO:0008995), 3'→5' exoribonuclease/polyribonucleotide nucleotidyltransferase (GO:0004654), ATP-dependent RNA helicase (GO:0004004), RNA pyrophosphohydrolase (GO:0034353-adjacent Nudix activity).

3. Expected step model (module) → status in KT2440

#	Module step	Expected player	KT2440 candidate	Status
1	Degradosome core assembly (scaffold + enolase)	RNase E family + enolase	<code>rne</code> PP_1905 (scaffold); <code>eno</code> PP_1612 (enolase)	covered (scaffold); candidate_uncertain (enolase membership)
2	Endonucleolytic initiation	RNase E catalytic	<code>rne</code> PP_1905	covered
3	ATP-dependent RNA unwinding	DEAD-box RhlB	<code>rhlB</code> PP_1295 (+ <code>rhlE</code> PP_4980, <code>rhlE-I</code> /RhpA PP_4766, <code>deaD</code> PP_1868)	covered but candidate_uncertain (paralog identity)
4	Phosphorolytic exonucleolysis	PNPase	<code>pnp</code> PP_4708 (± hydrolytic <code>rnr</code> PP_4880)	covered (PNPase present); module_needs_revision (RNase R may substitute)

4. Candidate genes and evidence

High-confidence degradosome-core genes

rne — **Ribonuclease E, PP_1905 (Q88LM4)**. *Role*: endoribonuclease + degradosome scaffold (CTD). *Evidence*: strong. Directly studied in KT2440 (RNase E deletion/variant among a panel of RNase mutants,

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Pseudomonas CTD scaffold function mapped in the close relative *P. aeruginosa* (

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Caveat: the Pseudomonas CTD is sequence-divergent from *E. coli*, so which SLiMs recruit which partners must be read from Pseudomonas data, not *E. coli* homology. **Promote to full review**.

pnp — **PNPase (polyribonucleotide nucleotidyltransferase), PP_4708 (Q88DW0)**. *Role*: 3'→5' phosphorolytic exoribonuclease; classic degradosome core partner and Pseudomonas CTD binder

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Evidence: strong; KT2440 pnp deletion characterized (

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Caveat: PNPase has well-documented degradosome-**independent** functions (e.g., cold/oxidative stress in related genera;

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so presence ≠ obligate degradosome participation. **Promote to full review**.

rh1B — **ATP-dependent RNA helicase Rh1B, PP_1295 (Q88NB7)**. *Role*: unwinds structured RNA to feed PNPase; Pseudomonas CTD binder

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Evidence: moderate (homology + *P. aeruginosa* interaction data). *Caveat*: **paralog ambiguity** — see below. **Promote to full review**.

eno — **Enolase, PP_1612 (Q88MF9)**. *Role*: glycolytic 2-phosphoglycerate dehydratase; in *E. coli* also a degradosome structural component

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Evidence for degradosome role in *Pseudomonas*: weak/absent — the *P. aeruginosa* CTD map named PNPase and RhlB (not enolase) as core partners

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Caveat: likely over-propagated as a "degradosome" annotation in this lineage; glycolytic annotation is solid. **candidate_uncertain.**

Accessory / neighboring RNA-metabolism genes (present, but not degradosome-core)

rnr — RNase R, PP_4880 (Q88DE6). Hydrolytic 3'→5' exoribonuclease for structured RNA.

Species-relevant twist: in *P. syringae* Lz4W, RNase R (not PNPase) co-purifies with RNase E and RhlE — a "degradosome of novel structure"

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KT2440 rnr deletion characterized

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Relevant to step 4; promote to full review to resolve whether it is a KT2440 degradosome partner.

rppH — RNA pyrophosphohydrolase, PP_5146 (Q88CN4). 5'-triphosphate→monophosphate

deprotection that licenses RNase E cleavage

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H. pylori ortholog

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Status: upstream **decay-initiation** step, adjacent to but outside the degradosome core. Homology-only in KT2440.

pcnB — Poly(A) polymerase I, PP_4697 (Q88DX1). Adds 3' poly(A) tails that mark RNA for

exonucleolytic decay. Accessory to 3'-decay; homology-based.

hfq — RNA-binding protein Hfq, PP_4894 (Q88DD3). In KT2440, characterized role is **Crc-**

Hfq catabolite repression (binds AAnAAnAA CA motifs, inhibits translation; regulates CrcZ/

CrcY sRNAs) —

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Its degradosome/decay role is not the primary demonstrated function here. **candidate_uncertain** for this module.

Helicase paralogs **deaD** PP_1868, **rhIE** PP_4980, **rhIE-I** /**RhpA** PP_4766. Additional DEAD-box helicases (EC 3.6.4.13). Create genuine ambiguity about which helicase serves the degradosome (RhlB vs RhlE, the latter favored in *Pseudomonas* per).

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Likely over-propagated / wrong-process bucket members

- **recQ** — DNA helicase **RecQ**, PP_4516 (EC 5.6.2.4). DNA repair/recombination. **Not an RNA-degradation enzyme; over-propagation into ppu03018.** → **not_expected_in_target_taxon** for the degradosome module (wrong process).
- **rho** — **Rho**, PP_5214. Transcription termination factor. Adjacent (transcription), not degradosome.
- **ppk** PP_5217, **ppkB** PP_0712. Polyphosphate kinases (KEGG "others" box). Peripheral.
- **groEL** PP_1361, **dnaK** PP_4727. Chaperones (primary bucket **ppu04156**). DnaK is only a stress-induced minor degradosome associate in *E. coli*, not a structural subunit

(P 16139413).

5. Gaps, ambiguities, and likely over-annotations

1. **Helicase identity gap (step 3).** RhlB is annotated, but *Pseudomonas* degradosomes can use RhlE

(P 15705581).

With four DEAD-box paralogs (**rhIB** , **rhIE** , **rhPA** , **deaD**), the actual KT2440 degradosome helicase is unresolved. → **candidate_uncertain** .

2. **Exonuclease substitution (step 4).** PNPase is present, but **RNase R** may substitute or co-participate in *Pseudomonas*

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The generic module's rigid "PNPase" wiring may under-represent KT2440. →

`module_needs_revision`.

3. **Enolase membership (step 1).** No KT2440 evidence that enolase is a structural degradosome subunit; homology transfer from *E. coli* is weak across *Pseudomonadota*. → `candidate_uncertain`; guard against auto-propagating a degradosome GO term to PP_1612.

4. **Bucket over-inclusion.** `recQ` (DNA), `rho` (transcription), `ppk` / `ppkB` (polyphosphate), `groEL` / `dnaK` (chaperones) inflate the module. Keep separate.

5. **Divergent physiology.** KT2440 RNase deletion phenotypes "diverged significantly" from *E. coli*

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— a general caution that *E. coli* functional annotations transfer imperfectly.

6. **RNase E regulators not captured by the bucket.** RraA-type regulators bind the RNase E regulatory domain and modulate its activity; an RraA-like protein exists in *P. putida* F1 (structurally characterized alongside HMG/CHA aldolases,

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These regulatory inputs are relevant to degradosome biology but are absent from the ppu03018 candidate list — a note for curators rather than a module step. (Not present as a KT2440 candidate gene; homology-level relevance only.)

6. Module and GO-curation recommendations

Module step / gene	Recommended mark	Rationale
Step 2 endonucleolytic initiation (<code>rne</code>)	covered	Present, essential, direct KT2440 + Pseudomonas scaffold data
Step 1 scaffold (<code>rne</code> CTD)	covered	Pseudomonas CTD scaffold demonstrated (P. aeruginosa)
Step 1 enolase (<code>eno</code>)	candidate_uncertain	Degradosome role homology-inferred only; weak transfer
Step 3 unwinding (<code>rh1B</code> / <code>rh1E</code> /...)	candidate_uncertain	Paralog identity unresolved; Rh1E plausible partner
Step 4 phosphorolysis (<code>pnp</code>)	covered	Present, KT2440-studied, Pseudomonas CTD binder
Step 4 wiring (PNPase-only)	module_needs_revision	RNase R substitution documented in Pseudomonas
<code>recQ</code> in this bucket	not_expected_in_target_taxon (wrong process)	DNA helicase; over-propagation
<code>rho</code> , <code>ppk</code> , <code>ppkB</code> , <code>groEL</code> , <code>dnaK</code>	keep in neighboring modules	Transcription / polyphosphate / chaperone processes
<code>rppH</code> , <code>pcnB</code> , <code>rnr</code> , <code>hfq</code>	accessory (candidate_uncertain for core)	5'-trigger / 3'-tailing / hydrolytic exo / regulator

Module-boundary recommendations: - The generic RNase E module is broadly correct but should (a) allow **Rh1E** as an alternative helicase player, (b) allow **RNase R** as an alternative/ added exonuclease player for Pseudomonads, and (c) demote enolase to an optional, lineage-variable structural participant. - Consider a **separate "5'-end-dependent decay initiation" module** (RppH) and a **"3'-end decay/polyadenylation" module** (PcnB + RNase R + PNPase), which the KEGG bucket currently conflates with the degradosome.

GO-curation notes: avoid propagating degradosome-complex GO terms onto `eno` (PP_1612), the chaperones, `recQ`, `rho`, and the polyphosphate kinases without direct evidence. A GO term / annotation request may be warranted to capture the *Pseudomonas*-specific **RNase R-containing degradosome** variant if curation needs to represent it.

7. Genes to promote to full `fetch-gene` review

1. `rne` PP_1905 — core scaffold + catalytic; confirm CTD SLiMs and partner set in KT2440.
2. `pnp` PP_4708 — core exonuclease; confirm degradosome vs independent roles.
3. `rh1B` PP_1295 and `rh1E` PP_4980 (and `rh1A` PP_4766) — resolve which helicase is the degradosome partner.
4. `rn1` PP_4880 — test for RNase E association (*Pseudomonas*-specific degradosome variant).
5. `eno` PP_1612 — verify/deny degradosome membership before annotation propagation.
6. (Secondary) `hf1` PP_4894 — clarify degradosome vs Crc-Hfq regulatory scope.

8. Key references

- `P 40096066`
(2025) Geslain et al. *Critical functions and key interactions mediated by the RNase E scaffolding domain in Pseudomonas aeruginosa*. — *Pseudomonas* RNase E CTD scaffold; SLiMs binding PNPase and Rh1B; membrane attachment; virulence/cold roles. (*Close relative; strong transfer to P. putida.*)
- `P 33089610`
(2021) Apura et al. *Ribonucleases control distinct traits of Pseudomonas putida lifestyle*. — **Direct KT2440:** PNPase, RNase R, RNase E, RNase III, RNase G deletions; roles diverge from *E. coli*.
- `P 15705581`
(2005) Purusharth et al. *Exoribonuclease R interacts with endoribonuclease E and an RNA*

helicase in Pseudomonas syringae Lz4W. — Pseudomonas degradosome with RNase R + RhlE (novel structure). (Same genus; moderate transfer.)

• [P 24803210](#)

(2015) Moreno et al. *The Crc and Hfq proteins of Pseudomonas putida cooperate in catabolite repression...* — **Direct KT2440** Hfq function.

• [P 27777366](#)

(2016) Hernández-Arranz et al. — KT2440 Hfq/Crc and CrcZ sRNA processing/stability.

• [P 37348756](#)

(2023) Lu et al. — Crc-Hfq as a portable translational-control module.

• [P 18202662](#)

(2008) Deana et al. *The bacterial enzyme RppH triggers mRNA degradation by 5' pyrophosphate removal.* — RppH as the 5'-deprotection trigger upstream of RNase E. (*E. coli*; mechanism generalizes.)

• [P 27974459](#)

(2017) Bischler et al. — RppH ortholog function in another Gram-negative (*H. pylori*).

• [P 23927922](#)

(2013) Domínguez-Malfavón et al. — *E. coli* degradosome assembly; RNase E-enolase and RNase E-PNPase interactions; CTD scaffold.

• [P 16139413](#)

(2006) Regonesi et al. — *E. coli* degradosome proteomics; DnaK a stress-induced minor associate, not structural.

• [P 23082859](#)

(2012) Henry et al. — PNPase degradosome-independent functions (*Yersinia*).

Evidence provenance summary

- **Direct KT2440 experiment:** RNase E, PNPase, RNase R, RNase III/G (deletion phenotypes); Hfq (Crc-Hfq).

- **Same-genus (Pseudomonas):** CTD scaffold/PNPase/RhlB (*P. aeruginosa*); RNase R + RhlE degradosome (*P. syringae*).
- **Homology / generic bacterial:** RhlB partner identity, enolase membership, RppH, PcnB, and all "others"-box genes.

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